Project Documentation

1. Introduction

- **Project Title:** AI Study Buddy API
- **Team ID:** SWTID-2026-2976
- **Team Size:** 4
- **Team Leader:** KARTHIKA P
- **Team member:** BHARATHIPRIYA N
- **Team member:** VAISHNAVI DEVI S
- **Team member:** ANANTHI S

2. Project Overview

- **Purpose:**
  AI Study Buddy is a smart web application designed to assist students in their learning process by providing personalized study support, doubt clarification, study material assistance, quizzes, and learning recommendations.
- **Features:**

- AI-based doubt clarification
- Study material and topic search
- Personalized learning recommendations
- Automatic notes and summary generation
- Quiz and practice question generation
- Question and answer assistance
- Student login and personalized dashboard
- Saved study materials and learning history

3. Architecture

- **Frontend:** React.js with Tailwind CSS / Bootstrap
- **Backend:** Node.js and Express.js for API and server logic
- **Database:** MongoDB to store student information, study materials, questions, quiz data, preferences, and learning history.
- **AI Component:** Artificial Intelligence / Natural Language Processing for understanding student questions and providing relevant learning assistance.

4. Setup Instructions

- **Prerequisites:**
  - Node.js
  - MongoDB
  - Git
  - React.js
  - Express.js
  - Visual Studio Code

●Installation Steps:

```
# Clone the repository
git clone <repository-link>

# Install client dependencies
cd client
npm install

# Install server dependencies
cd ../server
npm install
```

5. Folder Structure

```
AI-Study-Buddy/
|
|-- client/               # React frontend
|   |-- components/
|   |-- pages/
|
|-- server/               # Node.js backend
|   |-- routes/
|   |-- models/
|   |-- controllers/
|
|-- ai/                   # AI learning assistance
|   |-- services/
|   |-- prompts/
```

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6. Running the Application

●Frontend:

```
cd client
npm start
```

●Backend:

```
cd server
npm start
```

- **Access:** Visit:http://localhost:3000

7. API Documentation

- **User:**
  - "POST /api/user/register"
  - "POST /api/user/login"
- **Study Materials:**
  - "GET /api/study/search"
  - "POST /api/study/add"
  - "GET /api/study/:id"
- **Quiz:**
  - "POST /api/quiz/generate"
  - "GET /api/quiz/:id"
- **AI Assistant:**
  - "POST /api/ai/ask"
  - "POST /api/ai/summarize"
- **Learning History:**
  - "POST /api/history/add"
  - "GET /api/history/:userId"

8. Authentication

- JWT-based authentication for secure login.
- Middleware is used for protecting private routes.
- Students can securely access their personalized study information.

9. User Interface

The AI Study Buddy application contains the following main pages:

- Landing Page with study search and AI assistant
- Student Login / Registration Page
- AI Doubt Clarification Page
- Study Material Page
- AI Notes & Summary Page
- Quiz and Practice Page
- Student Dashboard
- Learning History Page

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10. Testing

Testing Methods:

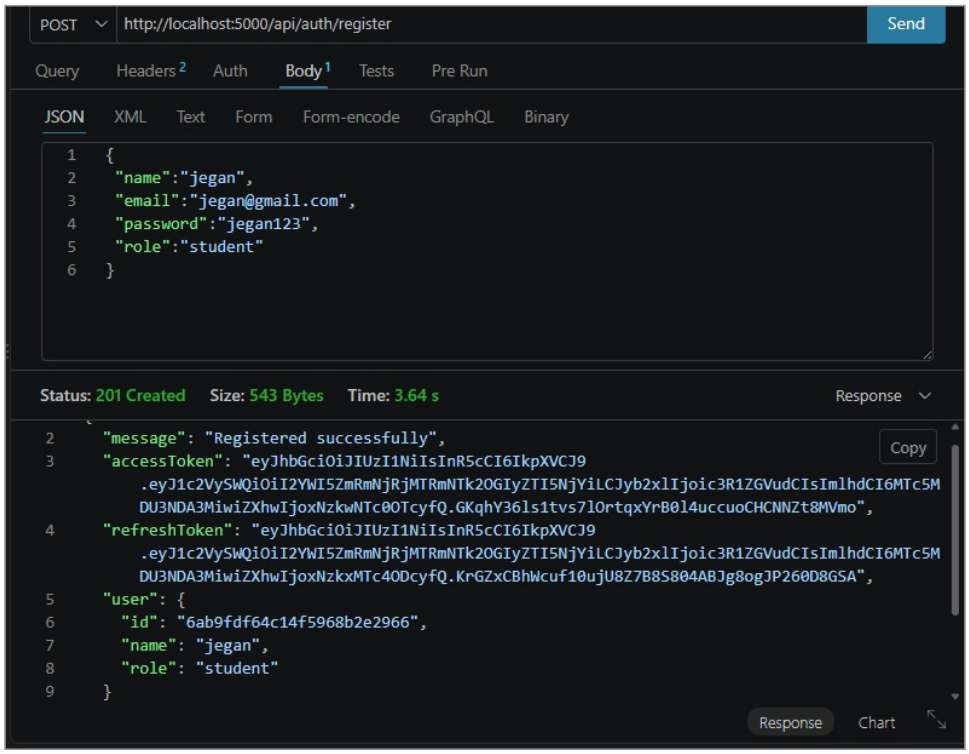
- Manual testing during project development

- API testing using Postman
- Browser testing using Chrome DevTools
- Unit testing using Jest

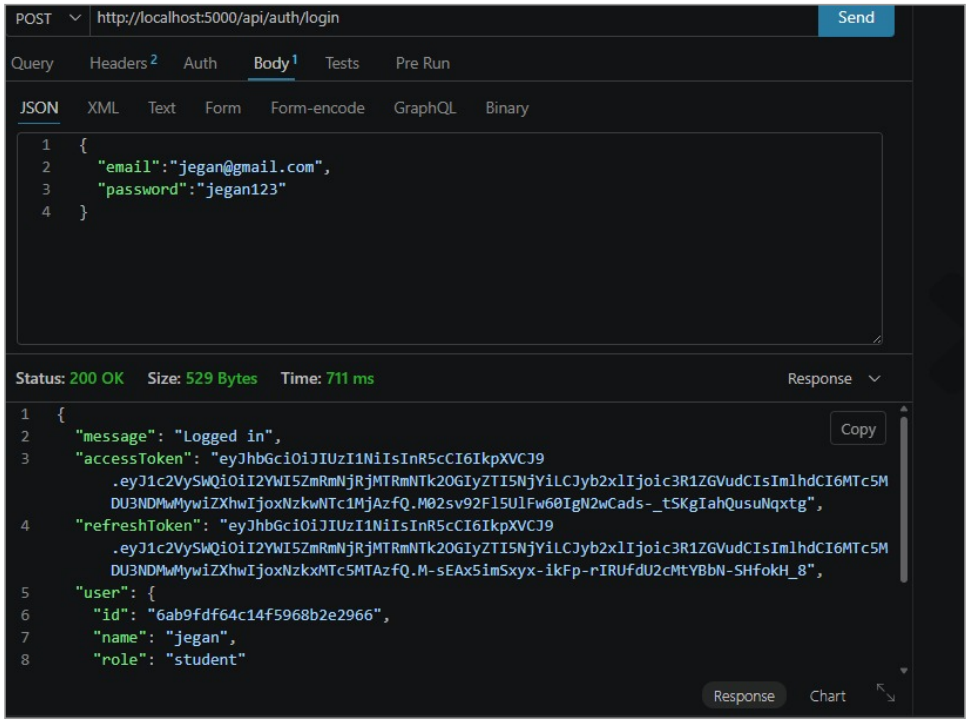
The application is tested to ensure that login, study material search, AI assistance, quiz generation, and other major functions work correctly.

11. Screenshots or Demo

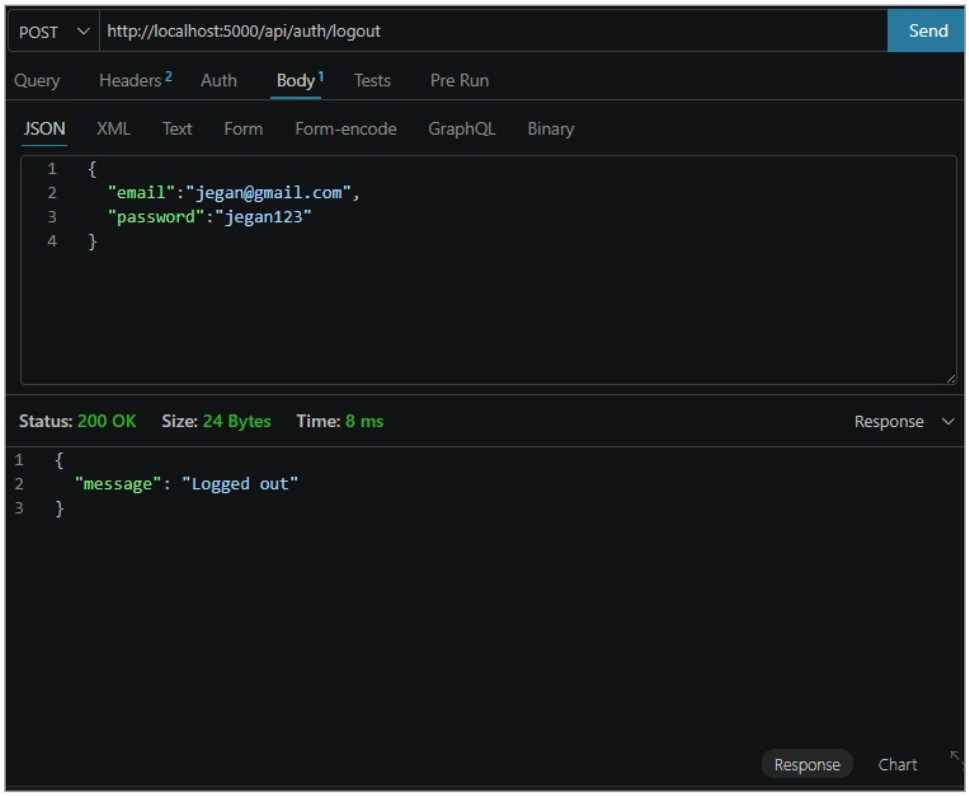
Step 1: Student Registration - POST /api/auth/register (201 Created)



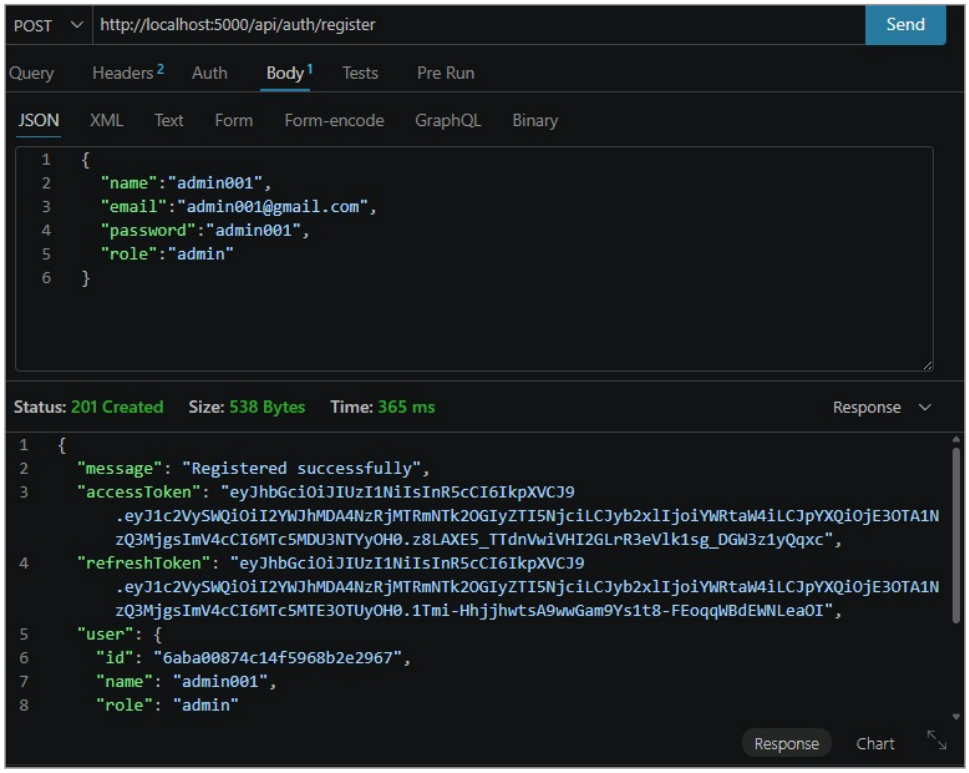
Step 2: Student Login - POST /api/auth/login (200 OK)



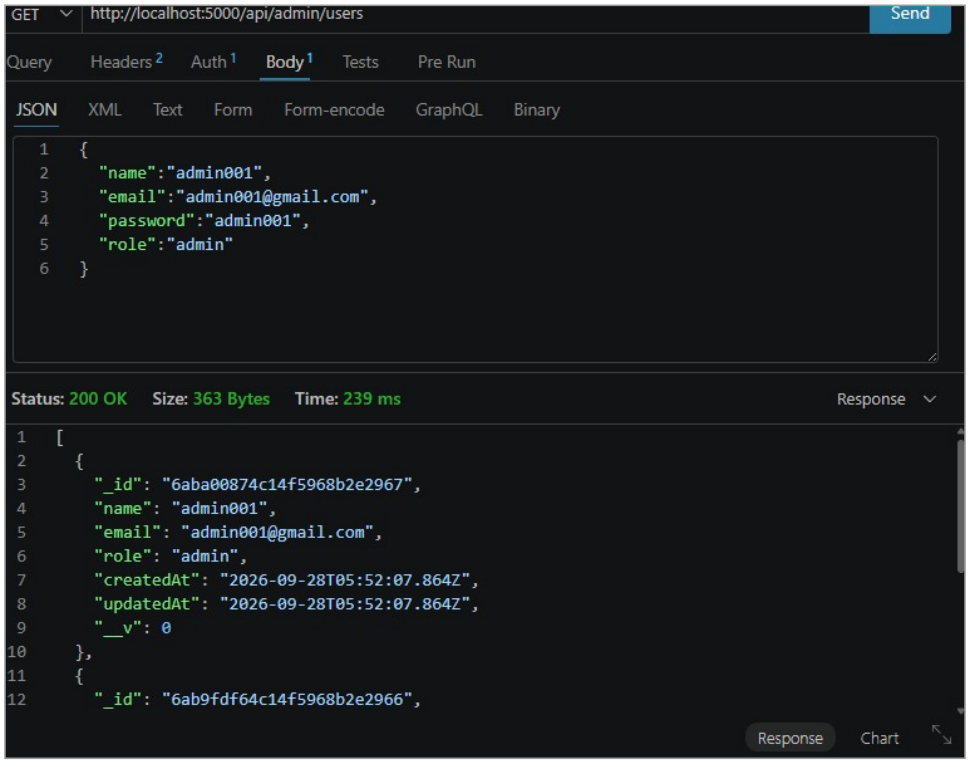
Step 3: Student Logout - POST /api/auth/logout (200 OK)



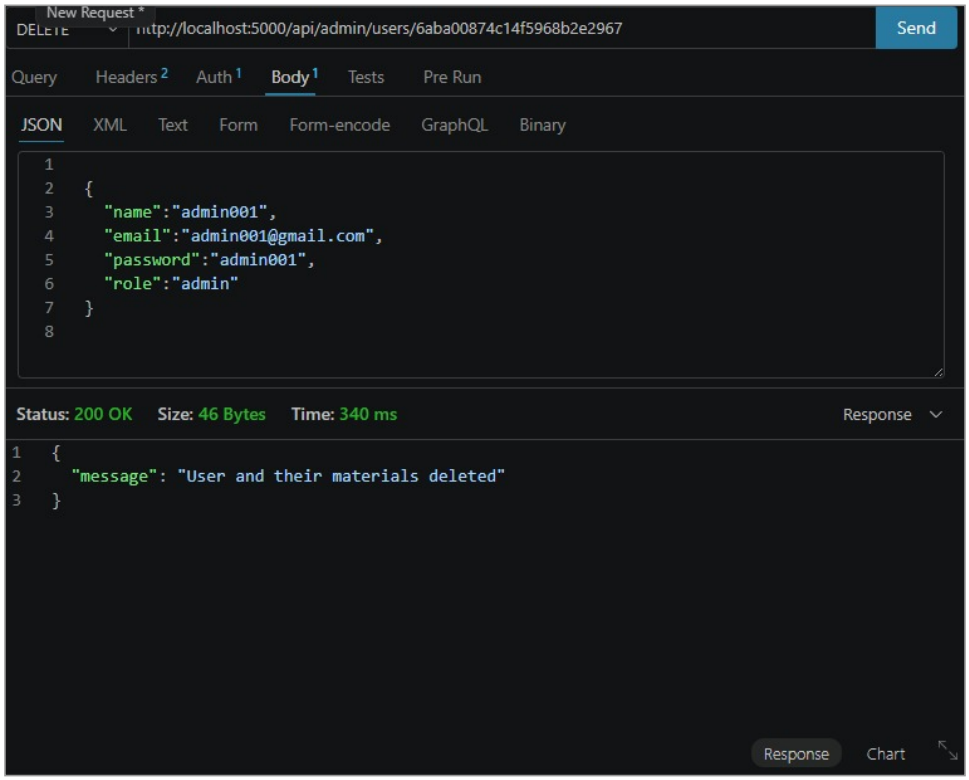
Step 4: Admin Registration - POST /api/auth/register (201 Created)



Step 5: Admin views all users - GET /api/admin/users (200 OK)



Step 6: Admin deletes a user - DELETE /api/admin/users/:id (200 OK)



12. Known Issues

- AI responses may vary depending on the question.
- Some complex topics may require additional study material.
- Internet connection may be required for AI-based features.

13. Future Enhancements

- Advanced AI-powered personalized learning system
- Multi-language support for study explanations
- Voice-based AI study assistant Automatic study timetable generation
- AI-based performance analysis
- Integration with online learning platforms
- Offline access to saved study materials
- Smart recommendation system based on student performance